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Dear Editor,

We are pleased to submit for your consideration the manuscript entitled “*A Unified Homotopy Framework for Nonlinear System Identification and Simulation: Constructive Dimensioning of Bilinear Neural Architectures*,” co-authored by Rodolfo H. Rodrigo and H. Daniel Patiño (Instituto de Automática, INAUT, UNSJ-CONICET).

The manuscript develops a unified algorithmic framework for two fundamental computational problems in nonlinear dynamical systems: parametric function approximation from sparse input–output data (system identification), and forward numerical integration of the resulting ordinary differential equation (simulation). Both tasks are reduced to closed-form Newton (z_1) and Halley (z_2) corrections derived from Liao’s zeroth-order homotopy deformation equation, exploiting the bilinear parameter structure shared by common function representations (radial basis functions, multilayer perceptrons).

The principal theoretical contribution is a *constructive dimensioning chain* that maps a target forward-simulation accuracy to approximation-theoretic bounds on the minimum number of basis functions, the minimum number of training samples (via Jacobian conditioning), and the maximum admissible integration time step (via Newton convergence on the discrete residual). The chain is constructive in the sense that each link is computable from system quantities, and it admits an operator-theoretic interpretation: the choice of the auxiliary linear operator $\mathcal{L}$ encodes prior knowledge that reduces the residual nonlinearity the approximator must learn, thereby reducing the required approximator dimension.

We validate the framework on three benchmark problems: a one-dimensional discharge-flow system (orifice-tank model), a two-dimensional Lotka-Volterra predator-prey system, and robustness experiments under varying levels of measurement noise. Small-scale parametric approximators (5–8 basis functions) identified from sparse data (20–40 samples) attain residuals below 10^{-6} in a single outer iteration and yield forward-simulation errors below 2% over 200-step horizons. We also provide extended wall-clock comparisons with modern gradient-based optimizers (Adam, L-BFGS) demonstrating the computational advantages of exploiting bilinear structure in small-network regimes.

The scope of the empirical validation is deliberately narrow—low-dimensional ODEs, small-scale approximators—and the role of the present work is to establish the algorithmic framework, the dimensioning theory, and the proof-of-concept computational regime. This positioning aligns naturally with the scope of *Applied Mathematics and Computation*: the contribution falls at the interface of approximation theory, numerical analysis for differential equations, and computational methods

for system identification, providing rigorous mathematical foundations with targeted empirical validation rather than large-scale engineering demonstrations.

We confirm that the manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. Companion manuscripts by the authors that share the same conceptual lineage but address distinct mechanisms (integral Volterra formulation; implicit-model sample complexity; coupled-system identification) are explicitly delineated in a dedicated section of the manuscript and are at different stages of preparation in venues that do not overlap with *Applied Mathematics and Computation*.

The reference implementations, benchmark scripts, and figure-generation code are publicly available at github.com/rodrigo1964-ai/unified-homotopy-framework and archived at Figshare (DOI: 10.6084/m9.figshare.31955865).

We thank you in advance for considering our work.

Yours sincerely,

Rodolfo H. Rodrigo
on behalf of the authors